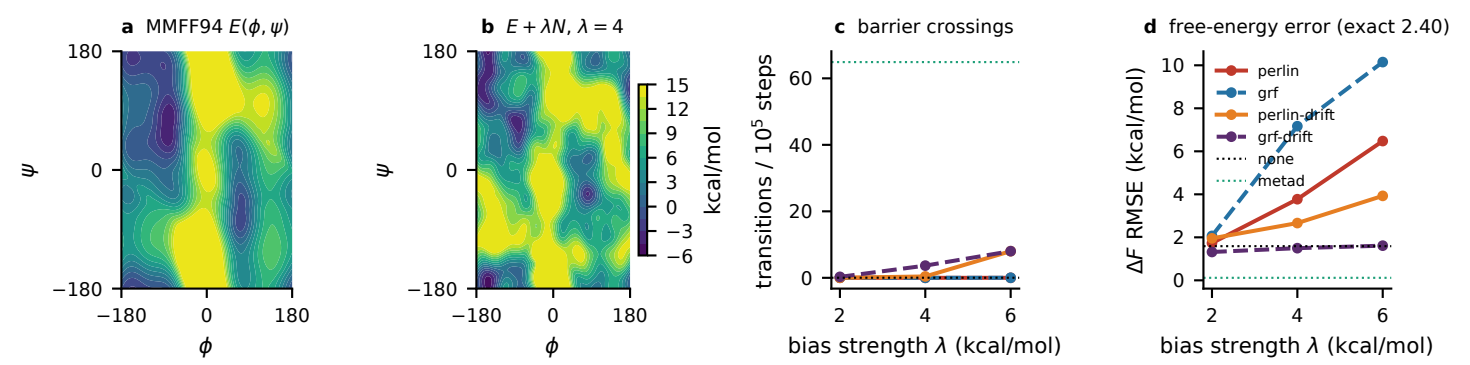

a MMFF94 $E(\phi, \psi)$
b $E + \lambda N$, $\lambda = 4$
c barrier crossings
d free-energy error (exact 2.40)
ψ
ϕ
180
0
−180
kcal/mol
15
12
9
6
3
0
−3
−6
transitions / 10^5 steps
bias strength λ (kcal/mol)
ΔF RMSE (kcal/mol)
perlin
grf
perlin-drift
grf-drift
none
metad